Name _____ _____ Baggett/Fall2026/207/Quiz3

first name last name

No Calculators.

Questions 1 and 2. Circle **True** or **False**. Then, in **one sentence**, say why — if the statement is false, the fastest way is usually to give an example that breaks it. **1 point for the circle, 2 for the reason.**

1. (**3** pts) A function can have two different horizontal asymptotes.

True **False**

Why? $\tan^{-1} x$ **has** $y = \frac{\pi}{2}$ **and** $y = -\frac{\pi}{2}$**.**

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2. (**3** pts) If $g(-2) = -5$ and $g(4) = 7$, then $g(c) = 0$ for some c in $(-2, 4)$.

True **False**

Why? Test the jump $g(x) = -5$ for $x < 1$ and $g(x) = 7$ for $x \geq 1$: it hits -5 at -2 and 7 at 4, but is never 0. The Intermediate Value Theorem requires g to be continuous on $[-2, 4]$, and nothing here says it is.

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3. (**4** pts) Evaluate the limit. **Show the step where the sign of x matters.**

$$\lim_{x \to -\infty} \frac{\sqrt{4x^2 + 3}}{x + 5}$$

-2

$$\lim_{x \to -\infty} \frac{\sqrt{4x^2 + 3}}{x + 5} = \lim_{x \to -\infty} \frac{\frac{\sqrt{4x^2 + 3}}{x}}{\frac{x+5}{x}} \qquad \text{divide top and bottom by } x$$

$$= \lim_{x \to -\infty} \frac{\frac{\sqrt{4x^2 + 3}}{-\sqrt{x^2}}}{1 + \frac{5}{x}} \qquad \textbf{here: } \frac{1}{x} = \frac{-1}{\sqrt{x^2}}, \text{ because } x < 0$$

$$= \lim_{x \to -\infty} \frac{-\sqrt{\frac{4x^2+3}{x^2}}}{1 + \frac{5}{x}} \qquad \text{one root over another is the root of the quotient}$$

$$= \lim_{x \to -\infty} \frac{-\sqrt{4 + \frac{3}{x^2}}}{1 + \frac{5}{x}} \qquad \text{divide inside}$$

$$= \frac{-\sqrt{4+0}}{1+0} = -2 \qquad \lim \tfrac{1}{x} = \lim \tfrac{1}{x^2} = 0$$

/4

4. (**5** pts) Let $f(x) = x^2 + 2x$.

(a) Find $f'(3)$ **from the definition**. Keep lim on every line until you actually take it.

$f'(3) = 8$

$$\begin{aligned} f'(3) &= \lim_{h\to 0} \frac{f(3+h) - f(3)}{h} \;=\; \lim_{h\to 0} \frac{\big[(3+h)^2 + 2(3+h)\big] - 15}{h} \\ &= \lim_{h\to 0} \frac{9 + 6h + h^2 + 6 + 2h - 15}{h} \;=\; \lim_{h\to 0} \frac{h^2 + 8h}{h} \\ &= \lim_{h\to 0} \frac{h(h+8)}{h} \;=\; \lim_{h\to 0} (h+8) \;=\; 8 \end{aligned}$$

(b) Write the equation of the tangent line to $y = f(x)$ at $x = 3$.

$y = 8x - 9$ **(slope** 8 **through** $(3, 15)$**)**

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5. (**3** pts) Show that the equation $x^3 + x = 1$ has a solution between 0 and 1. **Name the theorem you use, and check its hypothesis.**

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Let $f(x) = x^3 + x - 1$, so a solution of $x^3 + x = 1$ is a zero of f. f is a polynomial, so it is continuous on $[0, 1]$ — that is the hypothesis. Then

$$f(0) = -1 < 0 \qquad \text{and} \qquad f(1) = 1 > 0$$

so 0 lies between $f(0)$ and $f(1)$. By the **Intermediate Value Theorem** there is a c in $(0, 1)$ with $f(c) = 0$, and that c solves the equation.

6. (**2** pts) Write the equation of the horizontal asymptote of $f(x) = \dfrac{3x^2 - 1}{x^2 + 4}$.

$y = 3$

/2

Dividing by x^2 gives $\frac{3 - 1/x^2}{1 + 4/x^2} \to \frac{3}{1}$ as $x \to \pm\infty$, so the one horizontal asymptote is $y = 3$.